The preemptive breakup and extensive form games

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At the beginning of the *Seinfeld* episode, "The Pez Dispenser," Jerry and George have the following conversation in Jerry's apartment.

JERRY: I know, you told me you like her. Everything is going good.

GEORGE: No, everything is not going good. I'm very uncomfortable. I have no power. I mean, why should she have the upper hand?

Later, again in Jerry's apartment, George, Jerry, and Kramer come up with the idea of the *preemptive break up*.

GEORGE: Well it's over. It's definitely over.

JERRY: She broke up with you?

GEORGE: No, but I can tell she's going to. I can sense it. We had this terrible phone conversation. I was so nervous before I called I made up this whole list of things to talk about.

JERRY: What was on the list?

GEORGE: Let's see, how I'm very good at going in reverse in my car, why isn't Postum a more popular drink?

JERRY: Yeah, Postum is under-rated.

GEORGE: Anyway, there was all this tension. I asked her if she wanted to go out to dinner and she said no, maybe we could get together for lunch. You know what that means.

JERRY: What's wrong with lunch?

GEORGE: Lunch is fine at the beginning. Then you move on to dinner. You don't go back to lunch. It's like being demoted. I'll never do another crossword puzzle with her again. I know it.

KRAMER: I like the jumble You ever do the jumble?

GEORGE: I have no power. Don't you understand? I need [the upper] hand. …

KRAMER: Break up with her.

GEORGE: What?

KRAMER: You break up with her. You reverse everything that way.

JERRY: A preemptive breakup.

GEORGE: A preemptive breakup. This is an incredible idea. I got nothing to lose. We either break up, which she would do anyway, but at least I go out with some dignity. Or I completely turn the tables. This is absolutely brilliant.

What George has realized is that this is a two-player game and one of his options in this game is breaking up with his girlfriend, Noel, before she has a chance to break up with him. If he chooses that action, then either (a) she accepts the breakup, and, since he was the one who initiated it, he's able to "go out with some dignity" or (b) "I completely turn the tables" meaning that she resists breaking up, and so they remain together, but he now has the upper hand in the relationship (or at least he's in a better position than he was before).

If we try to set up this game in a table, however, we see that it doesn't work that well. If George initiates the break up, then Noel has the option to *accept the break up* or *try to keep the relationship intact*.

But if George doesn't breakup with her, then she has the option to *break up with him* or *not break up with him*.

		Noel			
		accept the breakup	resist the breakup	break up with George	don't break up with George
George	break up with Noel			?	?
	don't break up with Noel	?	?		

That, however, gives us four outcomes that can't happen—for instance, *he doesn't break up with her* and *she accepts the break up*. Those impossible "outcomes" are actually useful in more advanced game theory, but they prevent us from representing the game as it looks in reality. Plus, the table also leaves out an important feature of this game, namely, that George acts first and then Noel acts based on his choice.

The ***extensive form*** of a game represents actions that are taken in sequence rather than at the same time, and it does this by using a game tree instead of a table. (When games are represented in a table, they are in the ***normal form***.). The extensive form of this game is shown in figure 1. The points where a player has to choose an action are called ***choice nodes***, and the points where an outcome is reached are called ***terminal nodes***.

Since George selects his action first, he is at the top of the tree. Each of this options—*break up with Noel* and *don't break up with Noel*—presents her with two possible actions. If George tries to break up with her, then she can either *accept the breakup* or she can *reject the breakup*. If George doesn't break up with her, then she can *break up with him* or *not break up with him*. And George, to his credit, sees this entire game.

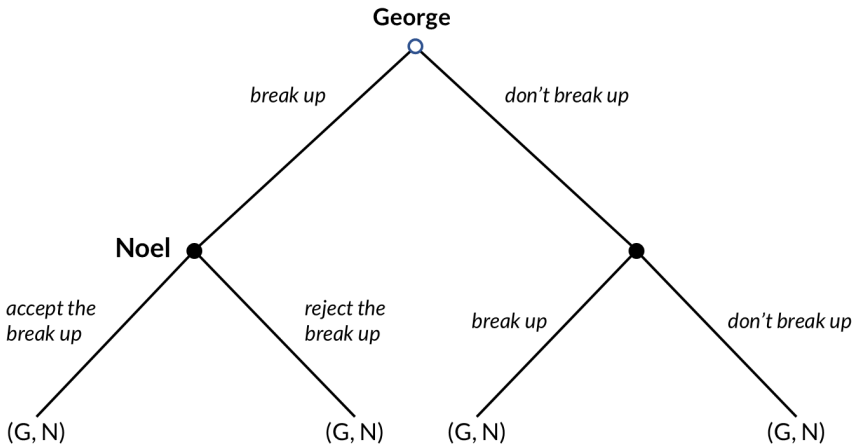


Figure 1: The extensive form of the preemptive break up. The 'G' and 'N' stand for George's and Noel's payoffs.

To find an equilibrium, we need to determine the value of each outcome for both players. We'll do this by ordering the outcomes from most to least preferred and then assigning utility numbers. George's most preferred outcome is *he breaks up with Noel* and *she rejects the breakup*. That involves her making some concessions and convincing him not to break up with her, which gives George the upper hand in the relationship. His next most preferred outcome, it seems, is *he doesn't break up with her* and *she doesn't break up with him*—although he thinks that this outcome is unlikely. Next best is *he breaks up with her* and *she accepts the break up*, and, last, *he doesn't break up with her* and *she breaks up with him*. That gives him this list with the accompanying utility numbers:

he breaks up with her, she rejects the breakup, 3

he doesn't break up with her, she doesn't break up with him, 2

he breaks up with her, she accepts the breakup, 1

he doesn't break up with her, she breaks up with him, 0

Meanwhile for Noel, since George believes she is going to break up with him, we'll assume that her two most preferred outcomes include breaking up. We'll also assume that, like George, if they are going to break up, she'd prefer to initiate it; and if they are not going to break up, she'd prefer not to have George try to break up with her. With that, these are her preferences (which also, by the way, make this a constant sum game):

he doesn't break up with her, she breaks up with him, 3

he breaks up with her, she accepts the break up, 2

he doesn't break up with her, she doesn't break up with him, 1

he breaks up with her, she rejects the break up, 0

Now we can complete the game tree. George's payoff is the first number in each pair, and Noel's is the second.

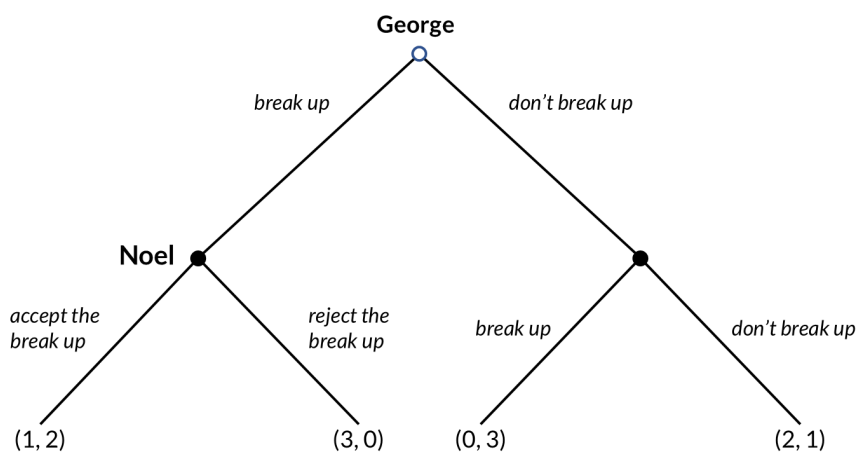


Figure 2 The first value in each pair of payoffs is George's payoff. The second is Noel's.

To solve this game, we'll use a procedure called *backward induction*. We first consider the action or actions that would be taken

by the player (in this case, Noel) whose choice nodes come right before the terminal nodes. Since Noel prefers 2 to 0, we know that, if George chooses *break up*, then she will choose *accept the break up*. Likewise, if George chooses *don't break up*, she will choose *break up* because she prefers 3 to 1.

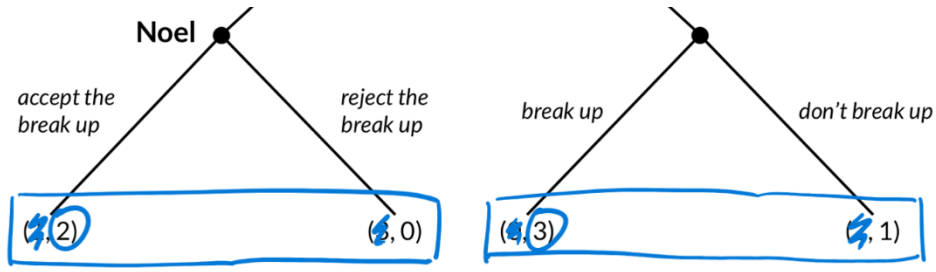


Figure 3 If Noel ends up at the choice node on the left, then she will choose *accept the break up* because it gives her a higher valued outcome than *reject the break up*.

If Noel ends up at the choice node on the right, then she will choose *break up* because it gives her a higher valued outcome than *don't break up*.

George knows this, and so his decision is, in effect, the one shown in figures 4 and 5. Since he prefers 1 to 0, he chooses *break up*. The outcome, then—and the solution to the game is—(*he breaks up with her*, *she accepts the breakup*), which has the payoff (1, 2).

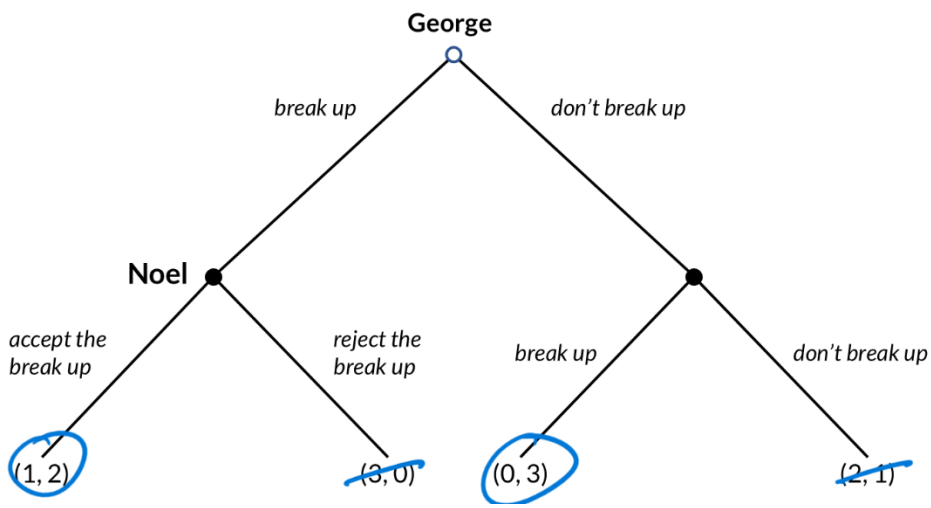
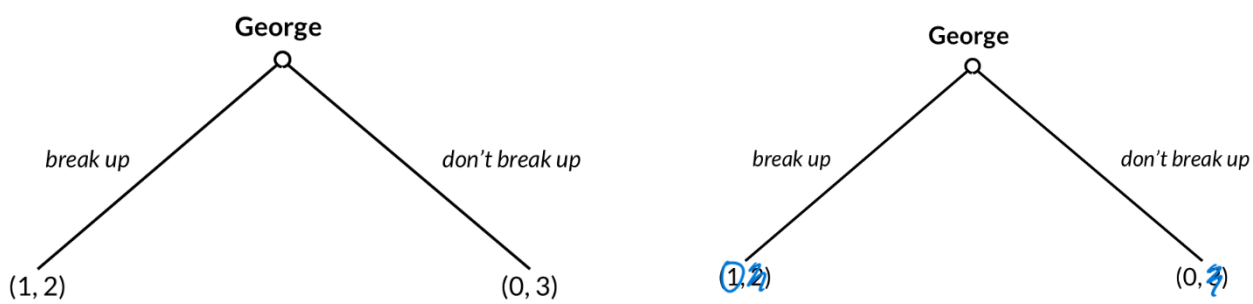


Figure 4 George can figure out that if he choose *break up*, then Noel will not choose the (3, 0) outcome when she can get (1, 2). And if he chooses *don't break up*, she will choose the (0, 3) outcome and not the (2, 1) one. Thus, his decision becomes the one in figure 5.



Figures 5a and 5b George's decision reduces to the one on the left. Since he prefers the payoff of 1 over the payoff of 0 (right), he will choose *break up*.

Another example

Consider another example of an extensive form game; one that was created by Tony Cookson and is shown in figure 6. First, Draco will cast spell *A* or spell *C*. Then Harry will cast spell *X* or *S* (in response to *A*) or *X′* or *S′* (in response to *C*).

If the strategy pairs (*A*, *X*) or (*C*, *X′*) are selected, then the game ends. If Harry selects *S* or *S′*, though, then Draco will have to choose between casting spells *a* or *c*.

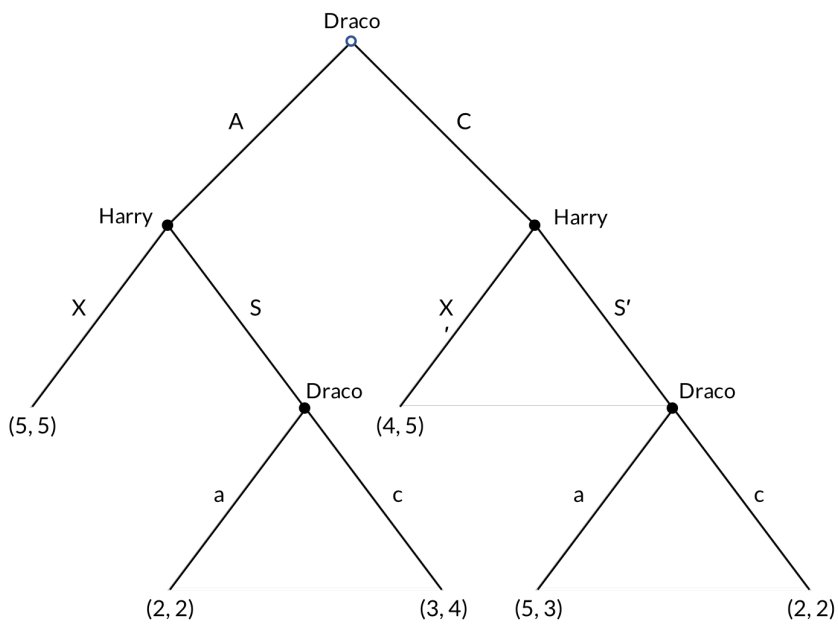


Figure 6

Starting at the bottom of the tree, Draco has two pairs of choices. Both are between casting spells *a* and *c*, but the outcomes depend, not only on the spell that he casts at that point, but on the spells that he and Harry have already cast. For the decision on the left (following *A*, *S*), Draco, who's payoffs are listed first, will choose *c*

because that gives him 3 instead of 2. For the decision on the right, he will choose *a* because that gives him 5 instead of 2.

That means that if Harry has to make a choice between *X* and *S*, he will choose *X* because he prefers 5 to 4. On the other hand, if he has to choose between *X′* and *S′*, he will choose *X′* because he prefers 5 to 3. Draco knows that Harry will make those choices, and so he knows that his choice between *A* and *C* will get him either 5 (if he chooses *A*) or 4 (if he chooses *C*). Hence, he will choose *A*. The outcome—and the solution—then, will be (*A*, *X*) and the payoffs will be (5, 5).[1]

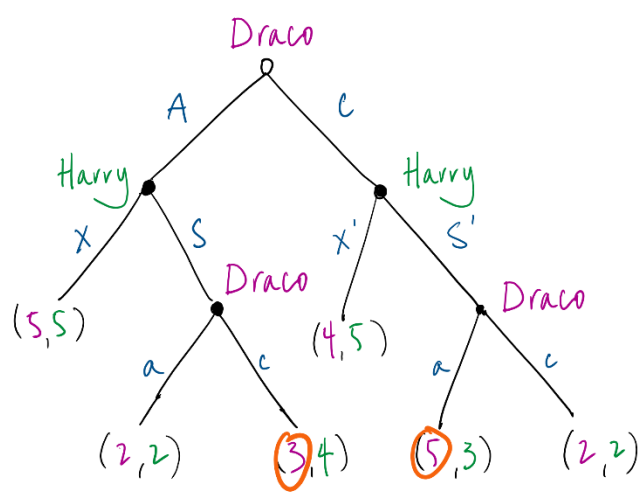


Figure 7

[1] We won't get into it, but just to flag it, in this game, defining each player's strategies (and the solution) is more complicated than just following the possible paths in the tree. Rather, all of the available information has to be mapped to the potential actions, even if some of those actions can never happen.

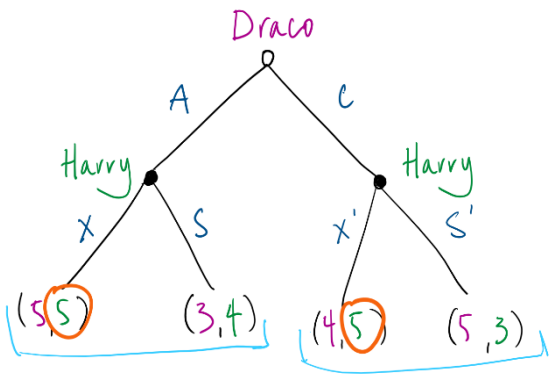


Figure 8 If Harry has to choose between X and S, he will choose X. And if he has to choose between X' and S', he will choose X'.

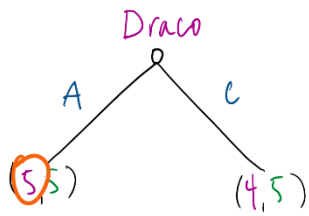


Figure 9 Knowing all of the information illustrated in figures 7 and 8, Draco will choose A to get the higher payoff (5 instead of 4).